Woojung Song

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RESEARCH INTERESTS

My long-term goal is to build agents that understand people and act in line with their goals and values.

My current research develops ways to evaluate the values and behavioral tendencies expressed by agents [ACL '25] [EMNLP '26] [arXiv '26]. I study how faithfully they simulate people across situations and over time [TACL '26] [EMNLP '26], and how to guide their behavior in interactions with users and tools [ICLR '26] [ACL '26] [arXiv '26].

EDUCATION

Seoul National University, Seoul, South Korea — Sep 2025 – Present
Integrated M.S.–Ph.D. in Data Science
Advisor: Yohan Jo, HOLI Lab (Human-Oriented Language Intelligence)

Hanyang University, Seoul, South Korea — Mar 2022 – Aug 2025
B.I.S. in Information Systems — GPA: 4.35/4.50 (overall and major)

PEER-REVIEWED PUBLICATIONS

* Equal contribution. Includes accepted papers.

ArcANE: Do Role-Playing Language Agents Stay in Character at the Right Time?
EMNLP, Main Conference (accepted) 2026
Woojung Song*, Nalim Kim*, Sangjun Song, Chaewon Heo, Jongwon Lim, Yohan Jo

Human Psychometric Questionnaires Mischaracterize LLM Behavior
EMNLP, Main Conference (accepted) 2026
Woojung Song*, Dongmin Choi*, Yoonah Park, Jongwook Han, Eun-Ju Lee, Yohan Jo

Don't Adapt Small Language Models for Tools; Adapt Tool Schemas to the Models
ACL, Main Conference 2026
Jonggeun Lee*, **Woojung Song***, Jongwook Han, Haesung Pyun, Yohan Jo

Non-Collaborative User Simulators for Tool Agents
International Conference on Learning Representations (ICLR) 2026
Jeonghoon Shim, **Woojung Song**, Cheyon Jin, Seungwon Kook, Yohan Jo

Psychometric Item Validation Using Virtual Respondents with Trait-Response Mediators
Transactions of the Association for Computational Linguistics (TACL) 2026
Sungjib Lim, **Woojung Song**, Eun-Ju Lee, Yohan Jo

Quantifying Data Contamination in Psychometric Evaluations of LLMs
Findings of EACL 2026
Jongwook Han*, **Woojung Song***, Jonggeun Lee*, Yohan Jo

Interpretable Prediction of Private Brand Purchases by Pet Type in E-Commerce for Consumer Behavior Analysis Using Real-World Transaction Data
PeerJ Computer Science 2026
Jaehyuk Lee*, **Woojung Song***, Jina Kim, Yoona Chung, Eunchan Kim

Value Portrait: Assessing Language Models' Values through Psychometrically and Ecologically Valid Items
ACL, Main Conference 2025
Jongwook Han*, Dongmin Choi*, **Woojung Song***, Eun-Ju Lee, Yohan Jo

PREPRINTS

* Equal contribution. All papers below are under review.

AgentHabit: Characterizing Distinct Behaviors of Agents on Everyday Tasks
arXiv:2609.32795 2026
Woojung Song*, Hoyeol Yang*, Jeonghoon Shim, Sungjib Lim, Jonggeun Lee, Yunho Choi, Yohan Jo

Agents' Overreliance on Unreliable Tools
arXiv:2609.05587 2026
Hoyeol Yang*, **Woojung Song***, Taewon Kim, Jonghyun Song, Seoyeon Park, Yohan Jo

Source Preference in the Wild: How LLM Agents Favor Items by Source, and How to Reduce It
arXiv:2610.03195 2026
Jonghyun Song*, Haewon Park*, Jeonghoon Shim*, **Woojung Song**, Yohan Jo

RESEARCH EXPERIENCE

Seoul National University, HOLI Lab — Seoul, South Korea
Graduate Research (Integrated M.S.–Ph.D.) — Sep 2025 – Present

- Profiled 18 models across 23 behavioral axes on 86 everyday tasks in AgentHabit.
- Evaluated tool overreliance, character development in role-playing agents (ArcANE), and whether psychometric questionnaires predict LLM behavior.

Seoul National University, HOLI Lab — Seoul, South Korea
Undergraduate Research Intern — Jul 2024 – Aug 2025

WORK EXPERIENCE

LG Electronics, H&A Sound/Image Intelligence R&D Lab — Seoul, South Korea
AI Research Intern — Mar 2024 – Jun 2024

- Improved LLM-based web question answering using retrieval-augmented generation and multi-agent approaches.
- Developed WAAG, an automated QA evaluation system with a 40% improvement in accuracy.
- Achieved the highest level in the LG Software Competency Certification Test.

Republic of Korea Army, KATUSA, U.S. Eighth Army — Pyeongtaek, South Korea
Sergeant (Senior KATUSA) — May 2020 – Nov 2021

- Completed mandatory military service as a Korean Augmentation to the United States Army (KATUSA) soldier.
- Received a Certificate of Appreciation for exceptional service.

SCHOLARSHIPS

Presidential Science Scholarship for Graduate Students — South Korea
Ministry of Science and ICT — 2026 – 2027